

Early Warning System for Patient Deterioration: Project Master Directory

Welcome to the Master Directory for this assignment. Because the deep learning models (especially the Transformers in Generation 3) take significant time to execute, the training loops have been pre-run.

This notebook serves as an index to the executed scripts, notebooks, and their outputs. Please click the links below to view the executed code and results for each generation.

1. Generation 1: Baseline DNN (Tabular Vital Signs)

In this phase, we established a baseline using a shallow Deep Neural Network trained purely on static vital signs.

- [View Generation 1 Code & Results \(PDF\)](#)
 - [View Generation 1 Script](#)
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2. Generation 2: Recurrent Sequence Models (LSTM, GRU, BiLSTM)

In this phase, we transitioned to sequential modeling, utilizing a 24-hour window of vital signs to capture the temporal trajectory of patient deterioration.

- [View Generation 2 Code & Results \(PDF\)](#)
 - [View Generation 2 Notebook](#)
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3. Generation 3: ClinicalBERT NLP (Unstructured Text Notes)

In this final phase, we incorporated unstructured clinical nursing notes using a pre-trained Transformer model (ClinicalBERT) to capture early warning signs present in clinical language.

- [View Generation 3 Code & Results \(PDF\)](#)
 - [View Generation 3 Notebook](#)
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4. Final Evaluation & Report

The comprehensive evaluation comparing all six architectures, including confusion matrices and performance tradeoffs, is documented in the final report.

- [View Final Report \(PDF\)](#)